

5. [Maximum mark: 7]

Consider the function $f(x) = \frac{(2x+a)^3}{(x+5)^2}$, where $x \neq -5$ and $a \in \mathbb{R}^+$.

- (a) Find an expression for $f'(x)$, in terms of a . [3]

When $x = 1$, the tangent to the graph of f makes an angle of 70° to the horizontal.

- (b) Find the two possible values of a . [4]

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